MILITARY STANDARD

DATA STORAGE TRANSPARENCIES FOR FACTORED TRANSPARENCY RADAR SIMULATORS

TO ALL HOLDERS OF MIL-STD-786(ONR/NTDC)

1. THE FOLLOWING PAGES OF MIL-STD-786(ONR/NTDC) HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
2	15 APRIL 1965	2	1 JUNE 1964
5	15 APRIL 1965	5	1 JUNE 1964
7	15 APRIL 1965	7	1 JUNE 1964
8	15 APRIL 1965	8	1 JUNE 1964

2. RETAIN THIS NOTICE AND INSERT BEFORE THE TABLE OF CONTENTS.

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3. Holders of MIL-STD-786(ONR/NTDC) will verify that page changes and additions indicated above have been entered and will destroy the previous notice (notice page only). The latest notice (notice page) will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

FSC 6940

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1. SCOPE

1.1 Scope.- This standard covers the configuration, density patterns, and dimensions for the photographic transparencies used as a map data storage device in the factored transparency system of radar simulation.

1.2 Classification.- Two configurations of factored transparencies are defined herein as follows:

1.2.1 Type I.- This transparency is a photographic image on a glass plate 16 by 20 by 1/4 inches, contains the elevation and reflectance images, and covers a geographic area of 510 by 1,315 nautical miles. It is detailed on figure 1.

1.2.2 Type II.- This transparency is a photographic image on a pair of glass plates, each 18 by 18 by 1/4 inches. One plate contains the elevation data image while the other contains the reflectance data image. The transparencies cover a geographic area of 1,250 by 1,250 nautical miles and are assembled in a fixed relationship by the use of a frame as detailed in figure 2.

2. REFERENCE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

DRAWINGS

U.S. Naval Training Device Center

640426	Type II Transparency Frame
640427	Type II Transparency Assy
640435	Spacer Flat
640436	Retainer, Film Plate
640437	Screw, Registration, Film Plate

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications.- The following document forms a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN STANDARDS ASSOCIATION

PH2.19-1959 American Standard for Diffuse Transmission Density

(Application for copies should be addressed to the American Standards Association, Inc., 10 East 40th Street, New York, N.Y., 10016)

3. DEFINITIONS

3.1 Terms used.- The following are definitions of terms used in this standard:

3.1.1 Transmittance.- Transmittance is the percent of light transmitted by a transparency, relative to the amount of light impinging on it and measured according to Standard ASA PH2.19-1959.

3.1.2 Reflectance.- Reflectance refers to the percent of energy reflected by an object relative to the amount of energy impinging on it. The term is used herein in connection with both the intensity of return energy of radar from targets and with the intensity of return of light from a pattern of gray tones.

3.1.3 Density.- Reflection or transmission density is the logarithm, to the base 10 ($\log_{10}$), of the reciprocal of the reflectance or transmittance.

3.1.4 Gamma.- The slope of the "straight" portion of a plot of photo material of density vs. $\log_{10}$ exposure producing it.

3.1.5 Sayce chart.- This chart is manufactured only by W. & L. E. Gurley of Troy, New York. Only the 1958 Gurley Master shall be used to produce the 300-line charts, occupying approximately 2-1/8 by 5-3/4 inches, for the purposes of this standard.

3.1.6 Edge gradient.- Edge gradient is defined as the function $\frac{G^2 X}{N}$ where $G = \triangle D/\triangle X$ (D is density, X is the measurement interval, and N is the number of measurements) greater than 5.

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4.3.5 Number assignment.- The transparency manufacturer shall apply to the U. S. Naval Training Device Center, Port Washington, N.Y., 11050, Attention Code 51, for the sequential number for any given transparency.

5. DETAIL REQUIREMENTS

5.1 Transparency dimensions.- The transparencies shall be as follows:

(a) Type I transparencies shall conform to the dimensions and configuration of figure 1

(b) Type II transparencies shall conform to the dimensions and configuration of figure 2

5.2 Simulated area.- The dimensions of figures 1 and 2 provide for a maximum geographic area, at a scale of 1:5,000,000, as follows:

(a) Type I--510 by 1,315 nautical miles

(b) Type II--1,250 by 1,250 nautical miles

5.3 Map projection.- The transparency and the master map shall be made using appropriate Lambert Conic Conformal Projection.

5.3.1 Register marks.- Register marks shall appear on all transparencies. Register marks for type I transparencies are detailed in figure 1, and for type II transparencies are detailed in figure 2.

5.3.2 Elevation-transmittance relationship.- The designation of the governing elevation-transmittance relationship shall be engraved on the transparency in accordance with figures 1 or 2 as appropriate.

5.4 Processing control.- Identical step wedges shall be placed on all transparencies. Type I transparency step wedges shall be in accordance with figure 1. Type II transparency step wedges shall be in accordance with figure 2. All step wedges shall contain a density identical to each elevation increment on the elevation image and each radar return code of the reflectance image.

5.4.1 Densities.- All densities shall be measured with a densitometer conforming to Standard ASA PH2.19-1959.

5.4.2 Gray scale control.- A copy of the laboratory characteristic curve and the corresponding reflectance-elevation control relationship, upon which the reflectance of each master map is based, shall be permanently attached to each master map. When a processing laboratory is required to copy an existing master map, the laboratory shall adjust its characteristic curve to duplicate the original characteristic curve as produced by the original laboratory.

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5.4.3 Emulsion-gray tone reproduction.- The emulsion used to image the master map shall be capable of the reproduction of gray values between 2 and 97 percent reflectance to a gamma or contrast function of 1, when measured from the characteristic curve data as specified by 4.1.1.

5.4.3.1 Emulsion resolution.- The emulsion shall be capable of reproducing 36 lines per millimeter at full amplitude when a sample is contact-printed from a Sayce resolution chart (see 3.1.5) using a diffuse light source so the density of the lines is 1.0 when the resultant print is measured with a microdensitometer reading linear density units and using a 5 by 150 μ aperture.

5.4.3.2 Emulsion granularity.- The standard deviation of its granularity (ΔD) shall be less than 0.0335 at 1.58 density and 0.0192 at 0.32 density when measured on a microdensitometer using a 48 μ aperture.

5.4.4 Elevation-transmittance slope accuracy.- The transmittance slope, plotted as specified in 4.1.3, shall fall within parallel lines separated by a value equal to 0.07 density. The slope shall not vary from the selected elevation-transmittance curve, in displacement or nonparallelism by more than a value equal to plus or minus 0.035 density.

5.4.5 Elevation-transmission values, accuracy.- The transmission value for any given assigned elevation shall have an accuracy of plus or minus one-half of the local elevation increment.

5.5 Register marks, position.- The position of the register marks shall be established within an accuracy of $\pm$ 500 scale feet. The register marks shall be noted with commensurate accuracy in seconds of latitude and longitude.

5.5.1 Transparency registration.- Ninety-five percent of all corresponding points on the reflectance and elevation transparencies shall register to within 0.00075 inch (300 feet at the transparency scale factor of 1:5,000,000). The remaining 5 percent shall register to within 0.0015 inch.

5.5.2 Type II transparency assembly.- The type II images shall be assembled in a frame as detailed in figure 2 and with the registration accuracy required by 5.5.1. The frame, necessary adhesive and potting compound shall be in accordance with drawings 640426, 640427, 640435, 640436 and 640437.

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5.6 Type I transparency positioning.- The transparency shall be positioned in the simulator according to the arrangement and dimensions of figure 3. The bearing surfaces shown in figure 3 shall be used in all cameras used to produce transparencies and shall also exist in the transparency frame or holder of all surfaces.

5.6.1 Transparency positioning adjustment.- The positioning arrangement shall be such that a given transparency, exposed in such a jig, may be returned to a similar jig so that any point on the transparency is returned to the same position, relative to the jig, within ± 0.0025 inch. The simulators shall contain a means to correct misplacement of the image within a maximum of ± 0.003 inch. The placement correction system of the simulators plus the jigs shall be such that any point on the transparency can be repositioned to an accuracy of ± 0.0005 inch.

5.7 Transparency flatness.- The glass used for transparencies shall have a flatness of 0.0005 inch per linear inch or an overall flatness of 0.00035 inch per linear inch multiplied by either dimension of the glass.

5.8 Transparency discontinuities.- If the final transparency is fabricated utilizing separate sections or map panels, the change in transmittance across the boundaries of these panels shall be within the transmittance tolerance of that particular gray scale in accordance with 5.10 and 5.11.

5.9 Radar detail.-

5.9.1 Category I.- The requirements specified in RM-380, U.S. Air Target Charts, series 200/100, including the new RM-280 Interim High/Low Altitude Amendment, and RM-381 and RM-382, Area Radar Prediction Analyses (ARPA) shall be used as a guide in determining the radar data to be encoded on the factored transparencies.

5.9.1.1 Cultural targets.- The description of radar return features contained in this section is based upon the publications cited in 5.9.1 and is included herein in connection with the delineation of the densities in 5.11 by which these features shall be represented under various circumstances.

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5.9.1.1.1 Populated places.- Populated places include urban areas and are defined as a concentration of structures designed and built for human occupancy or human occupational activities. These structures vary in size from a few buildings to metropolitan areas. All populated places shall be portrayed to reflect the physical shape of the radar significant features of the area as far as possible in accordance with the following criteria:

(a) The radar significant features of populated places, in accordance with ATC series 100/200. The minimum dimensions of a target return area shall be 250 by 250 feet.

(b) Any break in continuity of cultural build-up which is 400 feet or larger with a sigma of ± 50 feet measured on a root-mean-square (rms) basis.

(c) Landmark buildings, such as large sheds, roofed storage areas, tanks, tank farms, missile launch sights, and the like, which are radar significant according to surface material and predominant height analysis, and have a minimum size of 250 by 250 feet:

(1) All predicted returns coded for return value shall be shown in their proper return value as delineated on the ATC series 200 or other source data specified by the pertinent training device specification

(2) All visually significant isolated buildings indicated on ATC by a solid black symbol and annotated shall be shown as determined by an analysis of the descriptive annotation

(3) All visually significant isolated buildings indicated on ATC by a solid black symbol and identified, shall be shown as one level above the surrounding reflectances

(4) All visually significant isolated buildings indicated on ATC by a solid black symbol and not identified shall not be shown

(5) Where a solid black symbol on ATC is surrounded by a first intensity return it shall be rendered as a first intensity return and merged with the surrounding reflectances

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